Zecheng Zhou



University of Macau · PhD Student
B.Sc., School of Public Health, Xiamen University

Email: yc57636@um.edu.mo · Phone: 18505092356

Biophysical Modeling | Cardiac Electrophysiology | Epidemiology | Infectious Disease Modeling

Profile

PhD student at the University of Macau working on multi-scale computational modeling of physiological systems. My research integrates ODE-based electrophysiology, calcium-handling dynamics, and biomechanical models to study cardiac arrhythmia mechanisms and organ perfusion.

Previously trained in Preventive Healthcare at Xiamen University, with research experience in infectious disease modeling and chronic disease epidemiology.

Skilled in mechanistic modeling, parameter optimization, and large-scale simulation using MATLAB, Python, and C++. Experienced with statistical modeling approaches including logistic regression, chain binomial models, and PSM-DID.

Education

PhD Student — University of Macau, Macau SAR, China
Sep 2025 - Present

B.Sc. in Preventive Healthcare — Xiamen University, Fujian, China
Sep 2020 - Jun 2025 · GPA: 82.34 / 100 · CET-6: 504 · CET-4: 499

Technical Skills

- **Programming & Analysis:** R, Python, C, MATLAB, web-crawling (Python)
- **Statistical Software:** SPSS, Stata
- **Tools:** EndNote, Microsoft Office, GraphPad Prism
- **Methods:** Logistic regression, chain binomial models, infectious disease modeling, PSM-DID, basic metabolomics analysis

Research Projects

Biophysical Modeling of Cardiac Calcium Dynamics and Mechano-Electric Feedback

PhD Research Project · Aug 2025 - Present
University of Macau

- Developed a biophysically detailed cardiac cell model by coupling multiple ODE systems describing calcium handling, ion-channel dynamics, and active force generation.
- Built a coupled multi-physics model integrating intracellular Ca^{2+} dynamics, ion-channel electrophysiology, and sarcomere contraction mechanics.
- Implemented stretch-activated currents and mechanical feedback to study mechano-electric interactions in cardiomyocytes.
- Developed spatially heterogeneous calcium buffering models to investigate how subcellular Ca^{2+} regulation influences arrhythmogenic events.
- Designed simulation protocols for calcium waves and delayed afterdepolarizations (DADs) under different pacing frequencies.
- Performed global sensitivity analysis (Sobol method) to quantify parameter influence on calcium transients and arrhythmia triggers.

Results:

- Presentation accepted at the 17th World Congress on Computational Mechanics & 10th European Congress on Computational Methods in Applied Sciences and Engineering (WCCM-ECCOMAS 2026), Munich, Germany, Jul 2026.

Image-Based Modeling of Liver Perfusion and Metabolism

PhD Research Project · Aug 2025 - Present
University of Macau

- Developing a digital liver perfusion model combining medical imaging, vascular flow modeling, and metabolic simulation.
- Reconstructed 3D liver geometry and vascular structures from CT imaging data.
- Built multi-scale perfusion models describing blood flow through hepatic vascular networks.
- Coupled perfusion with ODE-based metabolic models (oxygen, glucose, lactate transport).
- Investigated temperature-dependent changes in vascular resistance and perfusion dynamics relevant to ex-vivo organ perfusion.
- Developed computational pipelines for simulation of transport and metabolic activity in liver tissue.

Relationship between dietary sialic acid intake and serum HbA1c

Project Leader · Jan 2024 - Present
Population-based cross-sectional study, Xiamen, China

- Conducted a cross-sectional analysis of the association between dietary sialic acid intake and HbA1c levels.
- Performed logistic regression and related statistical analyses using SPSS and R.
- Gained experience in nutrient analysis methods and English scientific writing.

Main results:

- Co-authored paper in *Scientific Reports* (JCR Q1): "Relationship between dietary sialic acid intake and serum HbA1c: a population-based study in Xiamen, China."
- Poster presented at the South China Public Health Conference & CHPAMS 2024 Wall Poster Showcase.

Sialic acid intervention, LDL sialylation, and prevention of atherosclerosis

Project Leader · Jul 2023 - Present

- Used different substrate combinations to study trans-sialidase (TS) reaction characteristics.
- Explored modification of LDL by exogenous sialic acids and their potential as TS substrates.

Gains: basic laboratory operations and use of laboratory equipment. Results: team won second and third prizes in Xiamen University's Basic Medical Innovative Research and Experimental Design Forum.

Assessment of aggregated HFMD outbreaks in Fujian Province based on a chain binomial model

Project Leader · Jul 2022 - Jun 2023

- Established a chain binomial model based on the natural history of HFMD virus.
- Fitted the model to aggregated outbreak data in Fujian Province to estimate transmission parameters and predict cluster outbreaks.

Gains: strengthened skills in R, Python, infectious disease modeling, experimental planning, and project submission. Results: developed an "Online Geocoding Software Based on R Language" at ShenSoft.

Temporal shifts in 24 notifiable infectious diseases in China before and during COVID-19

Team Member · May 2023

- Crawled provincial-level surveillance data for 24 notifiable diseases.
- Contributed to modeling and prediction for selected diseases using R and Python.

Results:

- Co-authored paper in *Nature Communications* (IF 14.7, JCR Q1).
- Research reported on Xiamen University Science and Technology Department webpages and other media.

Enhancing probiotic effects of *Bifidobacterium bifidum* and preparation of engineered strains

Team Member · Dec 2023 - Present

- Performed metabolomics data analysis and screening of engineered strains.
- Learned project filing, patent drafting, and scientific writing.

Results: awarded Outstanding Project in the 8th Annual Innovation and Entrepreneurship Conference of Xiamen University.

Impact of long-term care insurance pilot on catastrophic medical expenditures in China

Leader · Apr 2024 - Present

- Evaluated the impact of long-term care insurance (LTCI) on catastrophic medical expenditures of urban and rural residents.
- Applied propensity score matching difference-in-differences (PSM-DID) models using Stata.

Results: manuscript under revision for *China Health Statistics* (Peking University Core / Xiamen University First Class Core Journal).

Academic Competitions & Awards

- **National:** College Students' Innovative Entrepreneurial Training Plan Program - National Project.
- **National:** Tenth National Forum for Undergraduate Innovation Research & Experimental Design Competition - Gold Award and Best Manner Award.
- **National:** 14th MathorCup Maths Applied Challenge 2024 - Second Prize (National), Provincial First Prize, Fourth Place in East China Competition.
- **School:** Xiamen University Outstanding Undergraduate Thesis - Second Prize.
- **School:** First and Second National Forum for Undergraduate Innovation Research & Experimental Design Competition, Xiamen University - Second and Third Prizes.
- **School:** 2022 Outstanding Achievements in Practical Teaching of Ideological and Political Theory Courses, Xiamen University - Second Prize.
- **Xiamen University Innovation & Entrepreneurship Conference:** Outstanding Project Award (Bifidobacterium project).

Selected Publications & Presentations

- **Zhou, Z.**, Zhang, W., Wang, X. *et al.* Relationship between dietary sialic acid intake and serum HbA1c: a population-based study in Xiamen, China. *Scientific Reports*, 14, 23026 (2024).
- Li, K., Rui, J., Song, W., Luo, L., Zhao, Y., Qu, H., Liu, H., Wei, H., Zhang, R., Abudunaibi, B., Wang, Y., **Zhou, Z.**, Xiang, T., & Chen, T. Temporal shifts in 24 notifiable infectious diseases in China before and during the COVID-19 pandemic. *Nature Communications*, 15, 3891 (2024).
- Li, K., Wu, J., Zhang, R., Xie, Y., **Zhou, Z.**, Yin, Q., Chen, Q., Rui, J., Guan, X., Zhao, Z., & Chen, T. Global, regional, and national burdens of pertussis among adults: a systematic analysis of age-specific trends using Global Burden of Diseases 2021 data. *Infectious Diseases of Poverty*, 14, 85 (2025).
- **Zhou, Z.**, & Lei, C. Spatial remodeling of myolysis and troponin C buffering couples subcellular Ca^{2+} dynamics to contractile function in atrial fibrillation. Accepted conference presentation, WCCM-ECCOMAS 2026, MS341B: Imaging-Based Mathematical and Computational Modeling in Biomechanics II, Munich, Germany, Jul 2026.

Contact

Email: yc57636@um.edu.mo
Phone: 18505092356